PHASE 2 · PLANNING DOCUMENT

Clinic HMIS for Dr. Jawaria Rasheed

Workflow, open-source options (OpenEMR, OpenMRS, Bahmni) and a recommendation for a gynecology & IUI clinic in Bahawalpur.

What this document covers

- What an HMIS does in a single-doctor gynecology / fertility clinic
- The full patient workflow and the IUI-cycle workflow (diagrams)
- How OpenEMR, OpenMRS 3 and Bahmni work — architecture, strengths, limits
- Side-by-side comparison, plus a fourth option: a lightweight custom system
- Sample screen layouts for the clinic, and recommended next steps

All diagrams and screen layouts in this document are original illustrations. To see the real products, use the official demo links on the last page.

SECTION 1

Where the HMIS fits in the 4-phase project

The HMIS is the **central database** of the clinic. The website and the WhatsApp assistant put appointments *into* it; the dashboard reads data *out of* it.

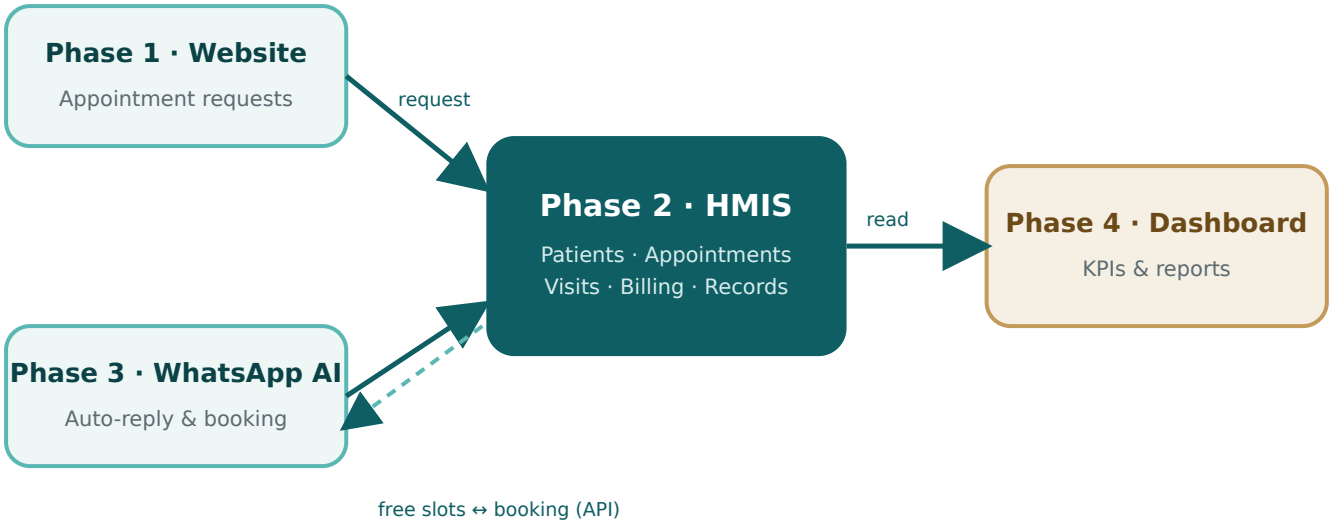


Figure 1 — Data flow between the four phases

What the clinic HMIS must handle

Module	Needed for Dr. Jawaria’s clinic
Patient registration	MR number, name, husband’s name, age, phone/WhatsApp, CNIC (optional), city, referral source
Appointments & queue	Doctor timings, slots, walk-ins, token number, status (booked / arrived / seen / no-show)
Consultation (OPD)	Complaints, LMP, obstetric history (G P A), vitals, examination, diagnosis (ICD-10), plan
Gynae / fertility records	ANC card, ultrasound findings, follicle-tracking chart, IUI cycle sheet, semen analysis, pregnancy outcome
Orders & results	Lab tests and scans ordered, results entered or attached (PDF/image)
Prescription	Medicine templates (e.g. letrozole protocol), printed prescription with clinic letterhead
Billing	Consultation, scan, IUI package fees, discounts, receipts, daily cash closing
Reminders	Follow-up and next-scan reminders by SMS / WhatsApp (Phase 3)
Security	Login per user, roles (doctor / receptionist / admin), audit log, daily backup

SECTION 2

Main patient workflow (OPD visit)

This is the day-to-day flow the HMIS must support, from booking to follow-up. Each box shows **who** does the step and **what is recorded**.

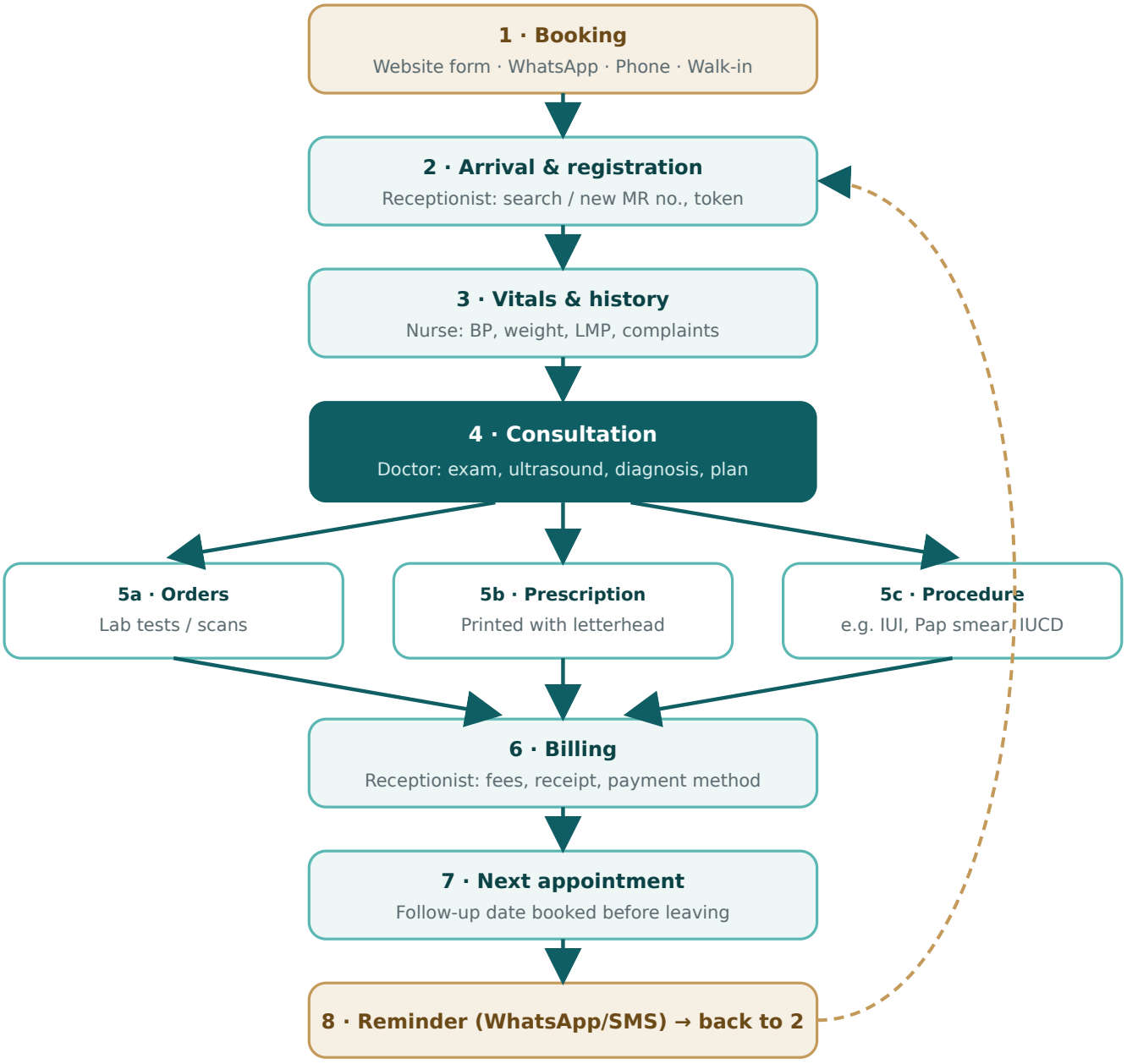


Figure 2 — OPD visit workflow

SECTION 3

Fertility & IUI cycle workflow

IUI is the clinic’s key specialty, so the HMIS needs a **cycle record** that follows one couple through several visits in the same menstrual cycle. None of the three open-source systems has this built in — it is added as a **custom form** in each of them.

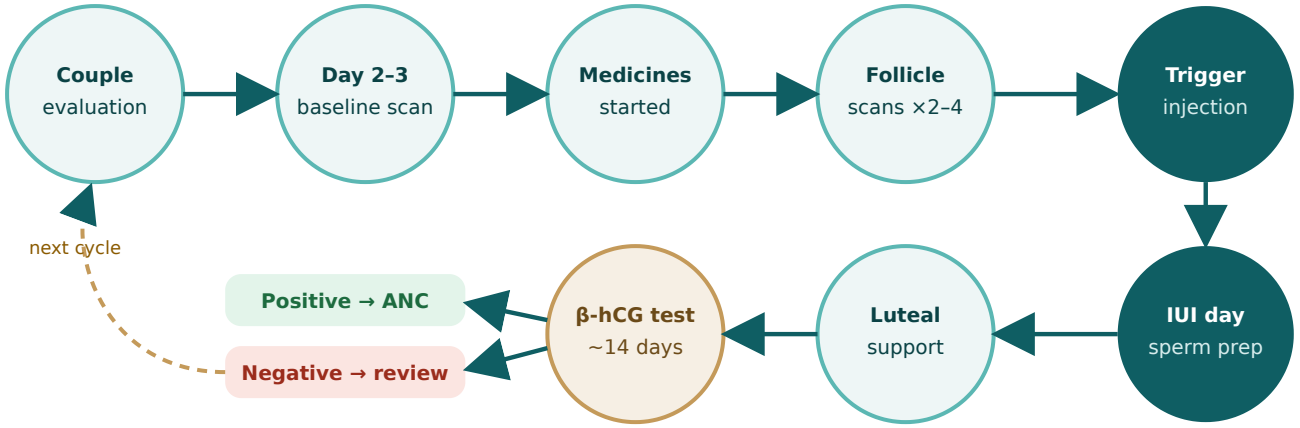


Figure 3 — One IUI cycle as recorded in the HMIS

Data captured in the IUI cycle sheet

Cycle details

- Cycle no., start date (LMP), protocol (natural / letrozole / injections)
- Medicines and doses per day
- Trigger date & time

Follicle tracking

- Scan date, cycle day
- Right / left ovary follicle sizes (mm)
- Endometrial thickness & pattern

Semen & procedure

- Pre-wash and post-wash count & motility
- IUI date/time, catheter, ease of procedure
- Doctor’s notes

Outcome

- β-hCG date & result
- Clinical pregnancy (scan) / miscarriage / live birth
- Feeds the Phase 4 dashboard (cycles & outcomes)

Outcome statistics should be shown only internally (dashboard) and only published on the website once verified and approved by Dr. Jawaria.

OPTION A

OpenEMR

A mature, free and open-source electronic medical record and practice-management system, used by clinics and small practices worldwide. Version 8.0 was released in February 2026.

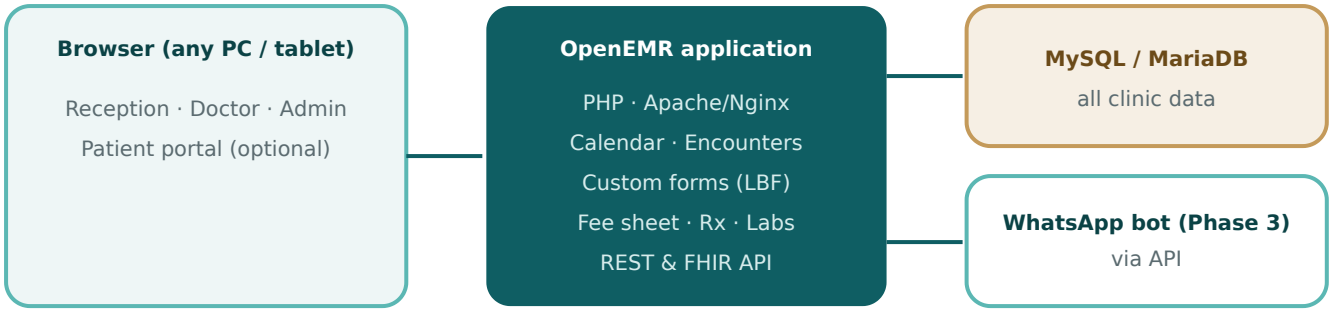


Figure 4 — OpenEMR architecture (simplified)

How the clinic workflow maps to OpenEMR

Workflow step	OpenEMR feature
Booking, queue	Calendar / appointment scheduler, patient flow board (arrived, in room, done)
Registration	Demographics screen (fields can be customised)
Consultation	Encounter with vitals, SOAP note, problem list, ICD-10 diagnoses
Gynae / IUI records	Layout-Based Forms (LBF) — build ANC card, follicle chart, IUI sheet without coding
Prescription	Prescription module with printable Rx
Billing	Fee sheet and cash receipts (US insurance features can be ignored)
Reports, API	Built-in reports; REST and FHIR APIs for WhatsApp bot and dashboard

✓ Strengths for this clinic

- PHP + MySQL — same stack as WordPress; easy for you to host and customise
- Designed for *clinics and practices*, not big hospitals
- Custom forms (LBF) for IUI and ANC without programming
- Large community, regular releases, APIs ready for Phase 3/4

✗ Limits

- Interface feels dated and busy; needs simplifying for reception staff
- Many US-specific features (insurance, US e-prescribing) to hide
- Pakistani billing/receipts need configuration or small customisation

OPTION B

OpenMRS 3 (O3)

An open-source medical record platform built by a global community for low- and middle-income countries. Version 3 (“O3”) has a modern interface built from small apps (micro-frontends).

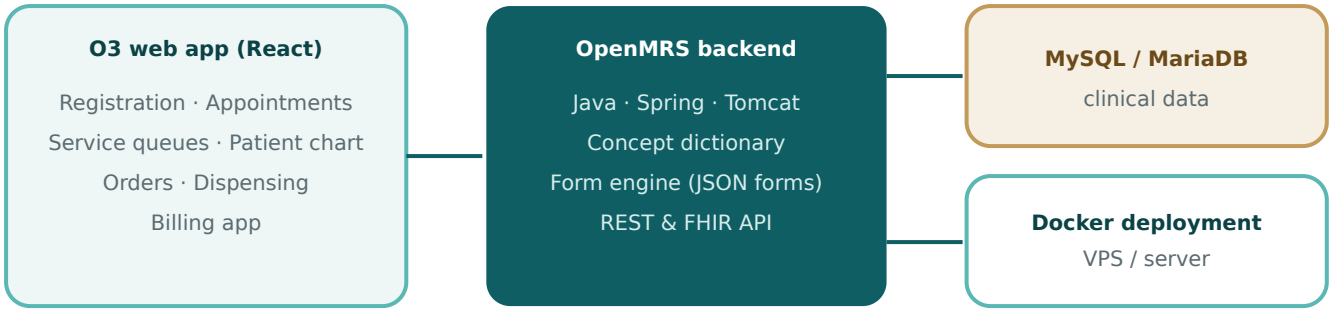


Figure 5 — OpenMRS 3 architecture (simplified)

How the clinic workflow maps to OpenMRS 3

Workflow step	OpenMRS 3 feature
Booking, queue	Appointments app and service queues (waiting → in consultation → done)
Registration	Patient registration app with configurable fields and ID generation
Consultation	Patient chart: vitals, visit note, conditions, diagnoses
Gynae / IUI records	Built with the form builder (JSON forms) after defining “concepts” (each data item)
Orders, Rx	Lab orders & results, medication orders, dispensing app
Billing	Billing app (basic); may need configuration for local fees

✔ Strengths

- Modern, clean interface; good on tablets
- Strong clinical data model and standards (FHIR)
- Built for low-resource settings; active global community

✘ Limits for this clinic

- Java stack — different from your PHP/WordPress skills
- Needs a VPS with Docker and more RAM than shared hosting
- Setting up concepts and forms has a learning curve
- Built for larger programs; a lot is unnecessary for one doctor

OPTION C

Bahmni

A complete open-source **hospital** system that combines several products into one: it uses OpenMRS for the medical record, and adds lab, billing/inventory and imaging. It comes in two editions — **Bahmni Lite** (for clinics) and **Bahmni Standard** (for hospitals).

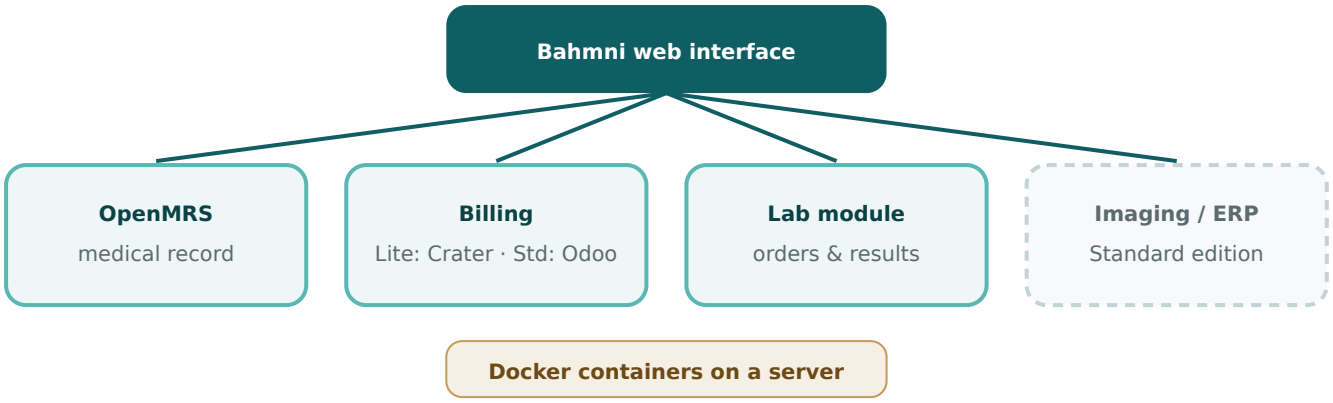


Figure 6 — Bahmni = several systems working together

Workflow step	Bahmni feature
Booking, queue	Appointment scheduling and registration desk
Consultation	Clinical dashboard, observations, diagnoses, medication
Gynae / IUI records	Custom observation forms (configuration)
Lab, billing	Integrated lab and billing (Lite: Crater; Standard: Odoo ERP with pharmacy stock)

✔ Strengths

- Most complete “all-in-one” package
- Proven in hospitals in low-resource countries
- Lab, billing and pharmacy connected out of the box

✘ Limits for this clinic

- Heaviest option: several systems to install, update and back up
- Needs a strong server (Docker, more RAM) and IT support
- Built for hospitals with wards and departments — far more than one gynecologist needs
- Customisation spans multiple products

SECTION 4

Side-by-side comparison

Option D is a fourth choice: a **lightweight custom clinic system** built only for Dr. Jawaria's workflow (PHP/Laravel + MySQL), connected to the WordPress website.

Criteria	A · OpenEMR	B · OpenMRS 3	C · Bahmni	D · Custom (light)
Best suited for	Clinics & practices	Programs, clinics, hospitals	Hospitals	One clinic, exact needs
Technology	PHP, MySQL	Java, MySQL, React	OpenMRS + billing + lab, Docker	PHP/Laravel, MySQL
Matches your skills	High	Medium	Low	High
Hosting needed	VPS (2 GB+) or good shared hosting	VPS with Docker (4 GB+)	VPS with Docker (8 GB+)	Same hosting as website or small VPS
Setup time (estimate)	2–4 weeks	4–8 weeks	6–10 weeks	6–10 weeks to build
IUI / gynae forms	LBF forms, no code	Form builder + concepts	Config forms	Built exactly as needed
Ease for receptionist	Medium	Good	Medium	Very good
API for WhatsApp (Phase 3)	REST + FHIR	REST + FHIR	Via OpenMRS API	Designed for it
Software cost	Free	Free	Free	Free (your build time)
Long-term maintenance	Community updates	Community updates	Several products to update	You maintain everything

Hosting sizes and times are planning estimates for a single-doctor clinic and depend on the amount of customisation and data migration.

Recommendation

1st choice — OpenEMR (Option A). It is made for clinics, uses PHP/MySQL like the website, gives scheduling, records, prescriptions, billing and APIs on day one, and the IUI and ANC sheets can be built as forms without programming. We simplify the screens for reception and hide the US-only features.

2nd choice — Custom light system (Option D) if, after trying the OpenEMR demo, it feels too complex for the staff. It takes longer to build but fits the clinic exactly.

Not recommended now: Bahmni (too heavy for one doctor) and OpenMRS (Java stack and setup effort higher than the benefit for this clinic).

SECTION 5

Sample screens for the clinic

Simple screen layouts showing what staff would see. These are design sketches (wireframes) to agree on before building, whichever option is chosen.

Today's appointments — Dr. Jawaria Rasheed — **Mon 26 Oct**

Token	Time	Patient (MR)	Reason	Source	Status
1	5:00	Patient A (MR-1021)	Follow-up · PCOS	WhatsApp	Seen
2	5:15	Patient B (MR-1088)	IUI cycle — scan day 10	Phone	With doctor
3	5:30	Patient C (new)	Infertility evaluation	Website	Arrived
4	5:45	Patient D (MR-0954)	ANC 28 weeks	Walk-in	Booked

+ New appointment | Register patient | Print list

IUI cycle sheet — Patient B · Cycle 2 — **Protocol: Letrozole**

LMP / Day 1	Medicine	Trigger	IUI date
12-10-2026	Letrozole 5 mg D3–D7	date / time	date / time

Scan date	Cycle day	Right ovary (mm)	Left ovary (mm)	Endometrium	Plan
14-10	3	—	—	4 mm	Start medicine
21-10	10	14, 11	9	7 mm	Repeat D12
23-10	12	18	10	8.5 mm, trilaminar	Trigger tonight

Semen — pre-wash	Semen — post-wash	β-hCG result
count / motility	count / motility	date / value

Save | Print sheet | Send reminder

Billing — receipt — **Cash closing: 26 Oct**

Service	Amount (PKR)	Paid by
Consultation · Follicle scan · IUI package	0,000	Cash / Card / Easypaisa / JazzCash

Save & print receipt | Daily cash report

Patient names and values above are placeholders, not real data.

SECTION 6

Next steps

#	Step	Who
1	Try the OpenEMR online demo (link below) as receptionist and as doctor; note what feels easy or difficult	Kamran
2	Confirm with Dr. Jawaria: paper forms she uses today (OPD card, ANC card, IUI sheet, Rx pad) — send photos	Kamran / Dr. Jawaria
3	Decide: OpenEMR (A) or custom light system (D)	Kamran
4	Install on your computer first (test), with sample data only	Claude + Kamran
5	Build forms: registration fields, OPD note, ANC card, follicle chart, IUI sheet, prescription templates, fee list	Claude + Kamran
6	Staff training (reception & doctor), 1–2 weeks trial alongside paper	Clinic
7	Go live on a secure server: HTTPS, user roles, daily off-site backup	Kamran
8	Connect the website appointment requests and plan the WhatsApp API (Phase 3)	Claude + Kamran

Privacy & security checklist (any option)

- Each staff member has their own login; doctor-only access to clinical notes
- HTTPS only; strong passwords and two-step login for admins
- Automatic daily backup stored off the server; test a restore monthly
- Audit log of who viewed or changed records
- Collect only data that is needed; patient consent for WhatsApp messages

Official links (see real screenshots and demos)

System	Link
OpenEMR	open-emr.org · Demo: open-emr.org/demo · Version 8.0 release (Feb 2026): open-emr.org/blog/openemr-version-8-released
OpenMRS 3	openmrs.org · Demo: o3.openmrs.org · Docs: openmrs.atlassian.net/wiki (O3 OpenMRS EMR)
Bahmni	bahmni.org · Releases & demo pages: bahmni.atlassian.net/wiki (Bahmni Lite Releases, Bahmni Online Demo)

Product details are summarised from the official websites and documentation as of October 2026. Features change between versions — confirm on the demo before deciding.